

Project Title
Community-Detection Platform

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

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ABSTRACT

Community-Detection Platform

The Community-Detection Platform is a graph-based analytical system designed to identify groups of closely connected users within large social, communication, or collaboration networks. The platform constructs a weighted interaction graph, where each node represents a user or entity and each edge represents the strength of interaction between two users. Edge weights can be determined using factors such as frequency of communication, common connections, and similarity in shared content.

Using graph-based community detection algorithms, the system partitions the network into meaningful communities whose members are more strongly connected to one another than to the rest of the network. The platform can support and compare multiple community detection approaches, allowing users to analyse how different algorithms perform on the same dataset. It also evaluates the quality of detected communities using graph metrics such as modularity, density, conductance, community size, and connectivity.

An interactive visualization module can display the network structure and highlight different communities, making complex relationships easier to understand. Users can explore individual nodes, connections, community statistics, and interaction patterns. The system may also include filtering and search features to analyse specific users or communities efficiently. The platform emphasizes the practical application of graph theory, data structures, algorithms, and performance analysis. It can be applied in social network analysis, recommendation systems, targeted digital marketing, collaboration analysis, fraud detection, and cybersecurity. Overall, the project provides an efficient and scalable approach for discovering hidden structures and meaningful relationships within complex network data.

Signature of the Guide